Milestone 2 Documentation — Bokone Community Network

Client Implementation Review, Feature Implementation and Test Evidence

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Client: Bokone Community Network (Vryburg)

Industry: Telecommunications

Client ID: CLI-148

Assigned Address Block: 172.30.100.0/23

1. Introduction

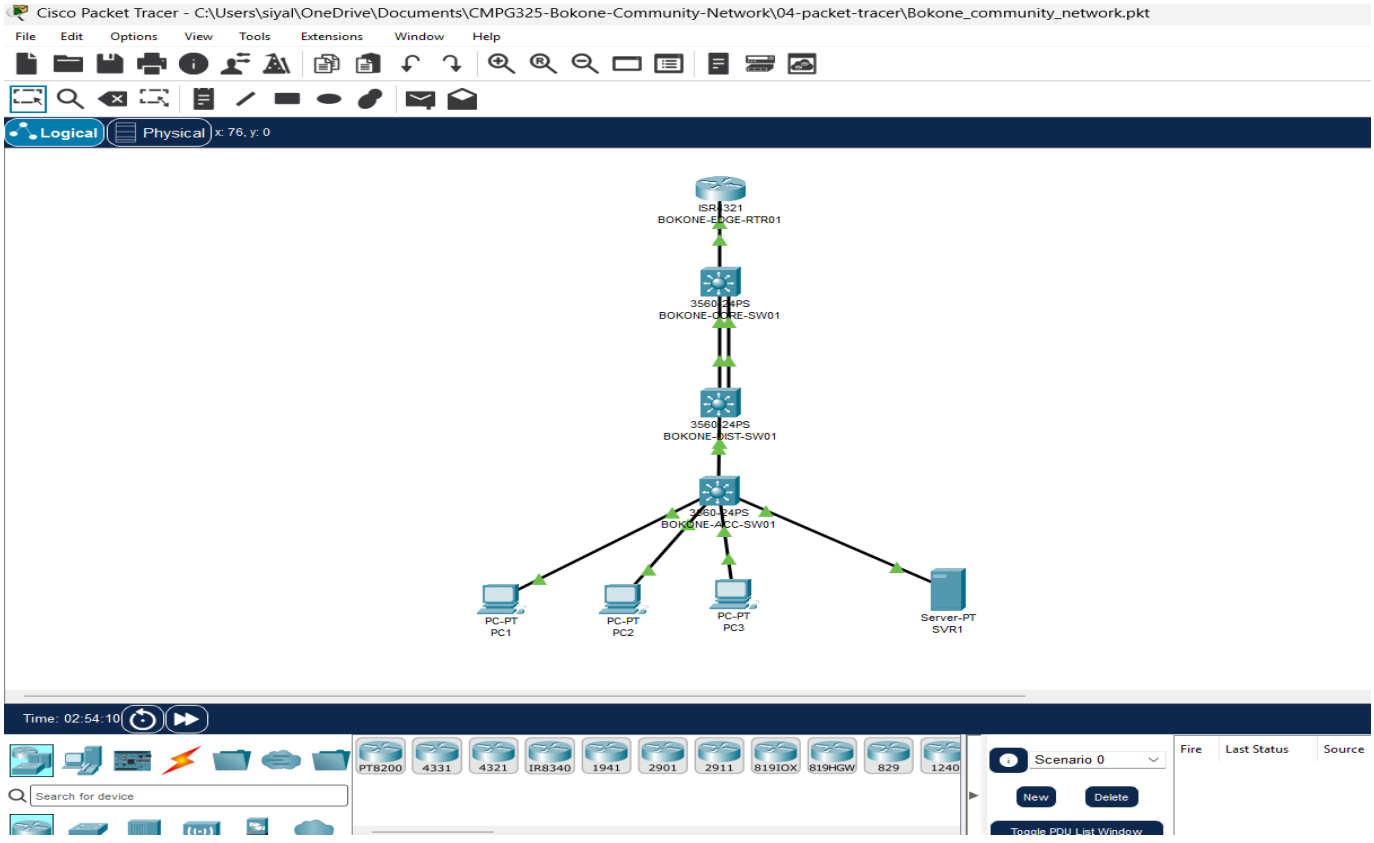
This document covers Milestone 2 of the Bokone Community Network project for CLI-148 (Bokone Community Network, Vryburg). The assigned technical challenge for this milestone was **EtherChannel**, used to combine two physical links between the core switch and the distribution switch into one logical, resilient connection.

This satisfies the client's requirement for a reliable switch-to-switch link that avoids a single point of failure and provides better bandwidth for traffic between organisational segments.

2. Network Topology

The network consists of four devices:

- **BOKONE-EDGE-RTR01:** router, connects the network to the outside (WAN) and routes to the core switch
- **BOKONE-CORE-SW01:** core switch, handles inter-VLAN routing (Layer 3) and is one end of the EtherChannel bundle
- **BOKONE-DIST-SW01:** distribution switch, the other end of the EtherChannel bundle, connects down to the access layer
- **BOKONE-ACC-SW01:** access switch, connects end devices (PCs and server) to the network



3. Device Configuration

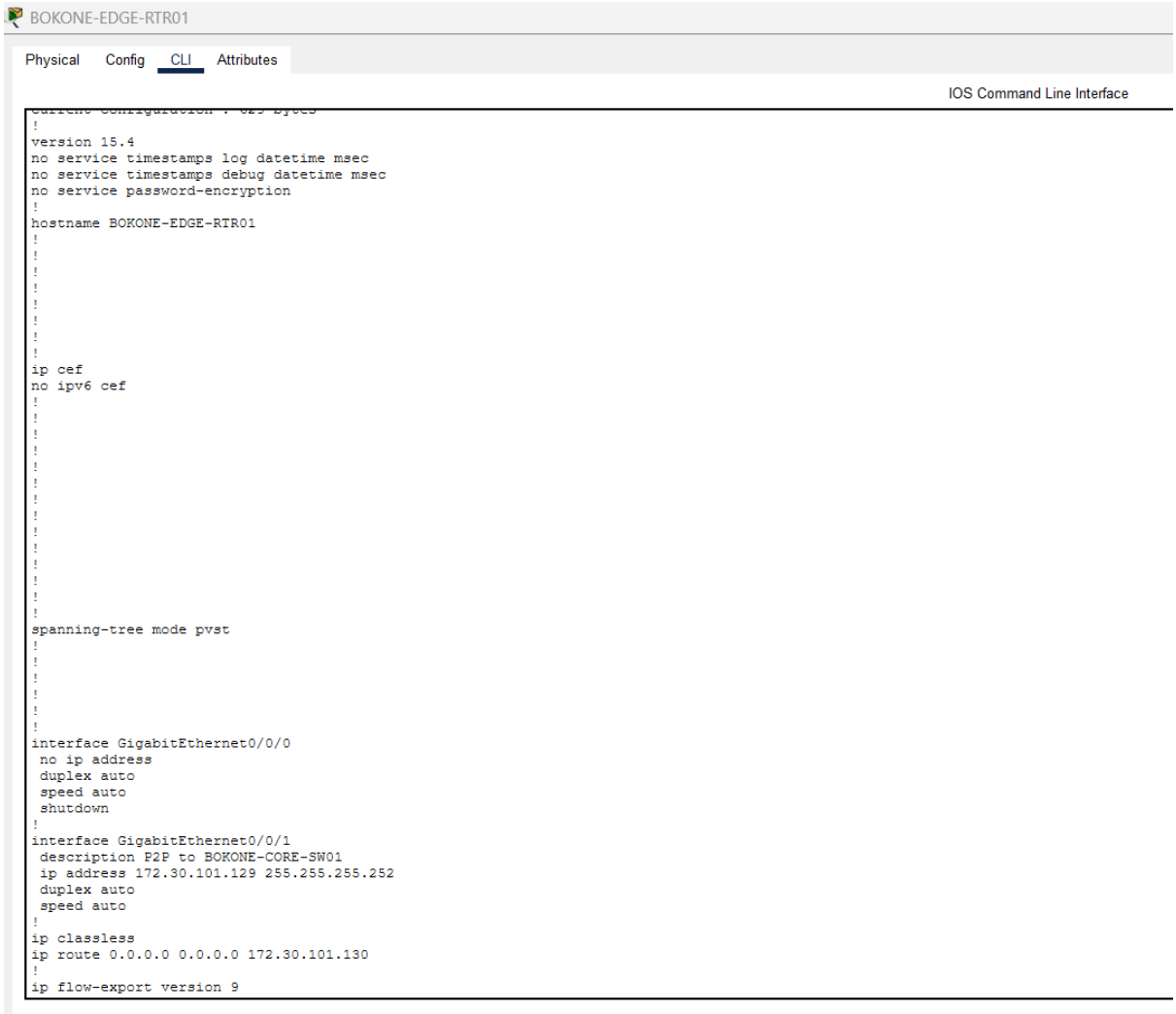
Below is what was configured on each device, with an explanation of why.

3.1 BOKONE-EDGE-RTR01 (Router)

The router connects to the core switch over a routed point-to-point link, using the subnet 172.30.101.128/30 reserved for this purpose in the IP Addressing Plan.

What was configured:

- Hostname set to BOKONE-EDGE-RTR01
- Interface GigabitEthernet0/0/1 given the IP address 172.30.101.129, connecting to the core switch
- This interface is the router's path into the rest of the network

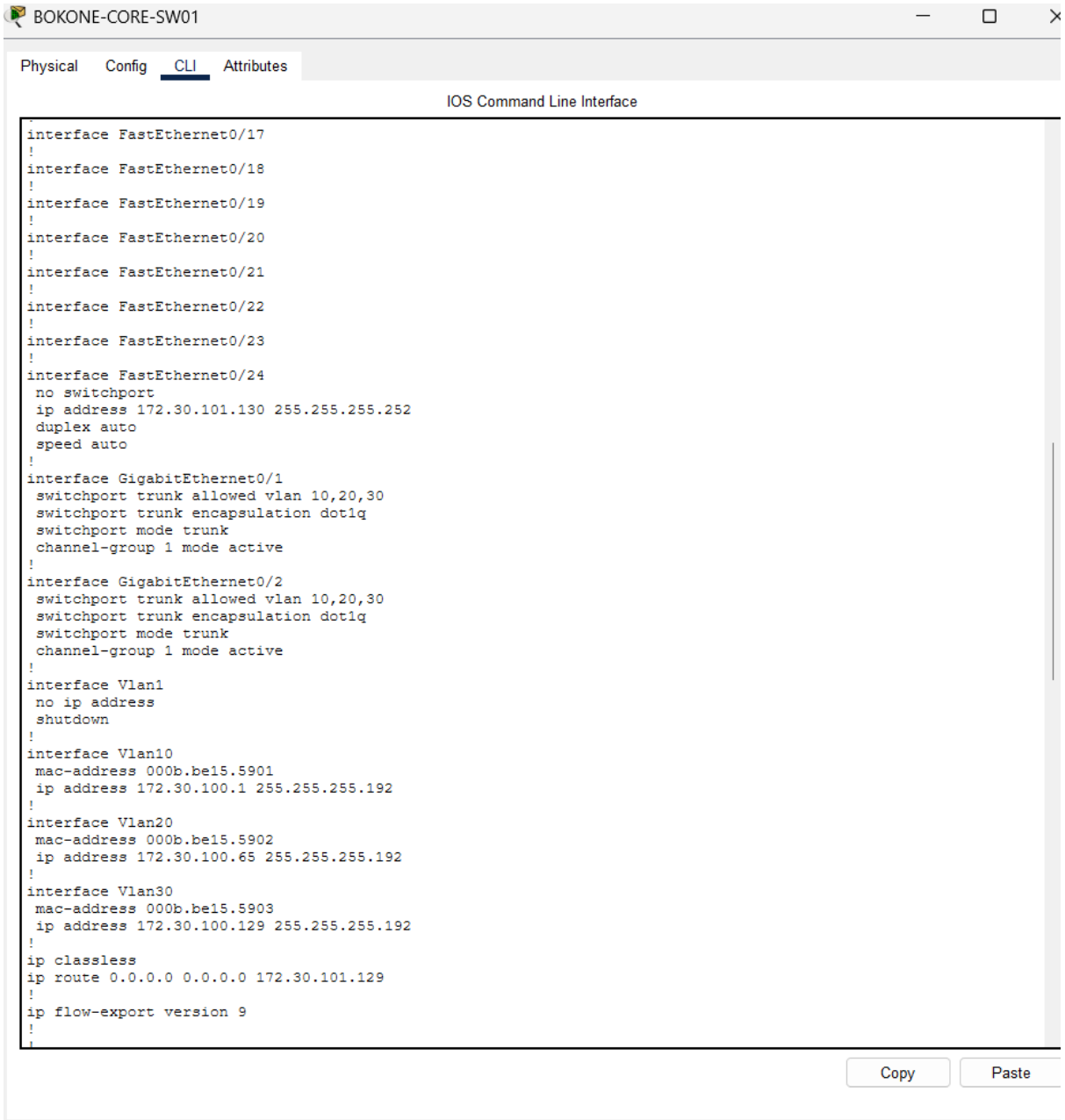


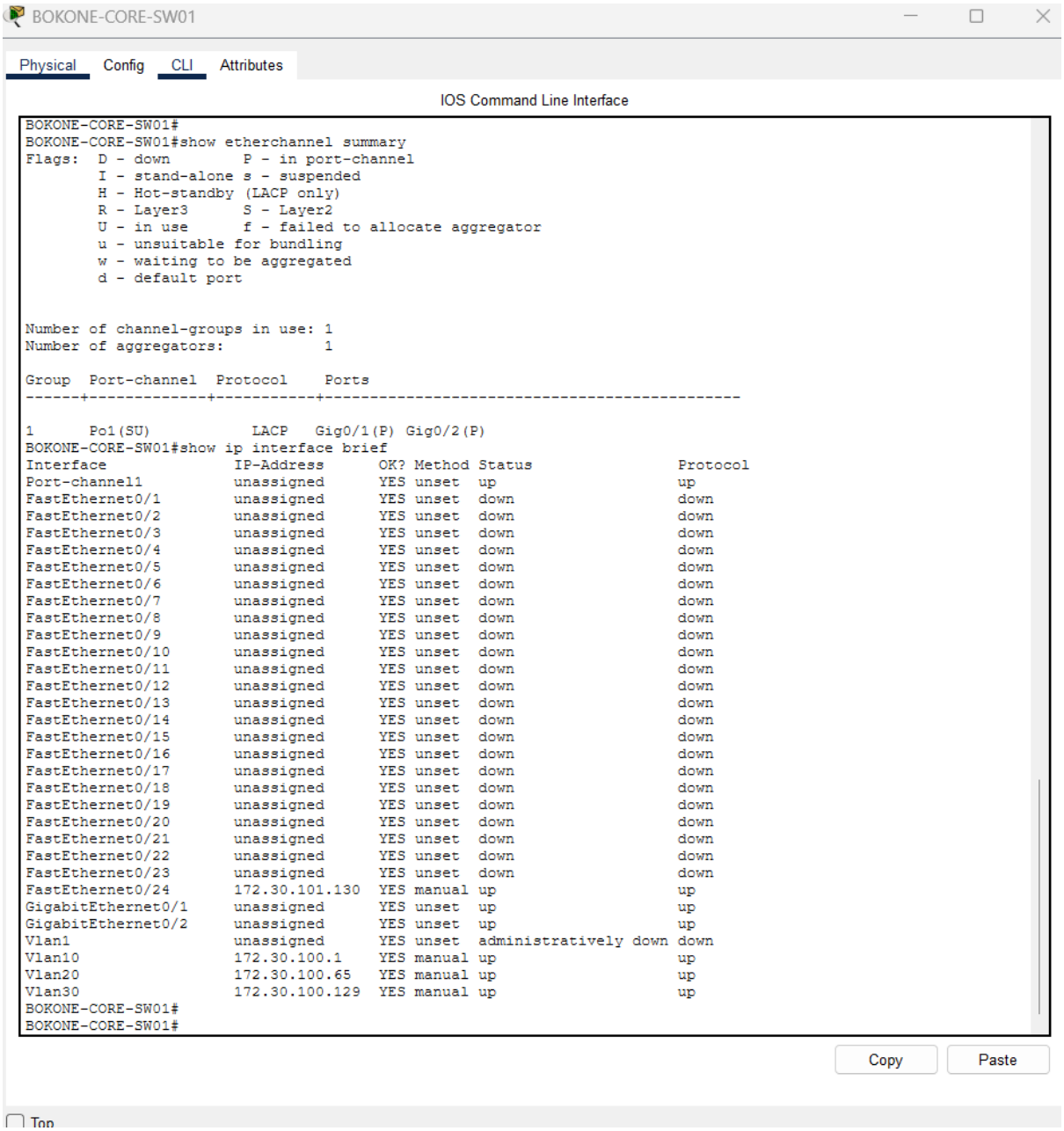
3.2 BOKONE-CORE-SW01 (Core Switch)

This is the most important device in the design it does two jobs: it routes traffic between VLANs, and it forms one end of the EtherChannel bundle.

What was configured:

- Hostname set to BOKONE-CORE-SW01
- IP routing enabled, so the switch can route between VLANs (not just switch within them)
- Three VLANs created: VLAN 10 (MGMT), VLAN 20 (STAFF), VLAN 30 (NOC_SRV)
- GigabitEthernet0/1 and GigabitEthernet0/2 configured as trunk ports, carrying all three VLANs, and bundled together into an EtherChannel using LACP (channel-group 1 mode active)
- A Layer 3 (routed) port, FastEthernet0/24, connects to the router with IP 172.30.101.130
- Three SVIs (virtual routing interfaces) created, one per VLAN, acting as the default gateway for each VLAN:
    - VLAN 10 gateway: 172.30.100.1
    - VLAN 20 gateway: 172.30.100.65
    - VLAN 30 gateway: 172.30.100.129
- DHCP configured for VLAN 20 (STAFF), so staff PCs get an IP address automatically
- A default route pointing back to the router, so traffic destined outside the network knows where to go


BOKONE-CORE-SW01
Physical
Config
CLI
Attributes
IOS Command Line Interface
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
 no switchport
 ip address 172.30.101.130 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 switchport trunk allowed vlan 10,20,30
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface GigabitEthernet0/2
 switchport trunk allowed vlan 10,20,30
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan10
 mac-address 000b.be15.5901
 ip address 172.30.100.1 255.255.255.192
!
interface Vlan20
 mac-address 000b.be15.5902
 ip address 172.30.100.65 255.255.255.192
!
interface Vlan30
 mac-address 000b.be15.5903
 ip address 172.30.100.129 255.255.255.192
!
ip classless
ip route 0.0.0.0 0.0.0.0 172.30.101.129
!
ip flow-export version 9
!
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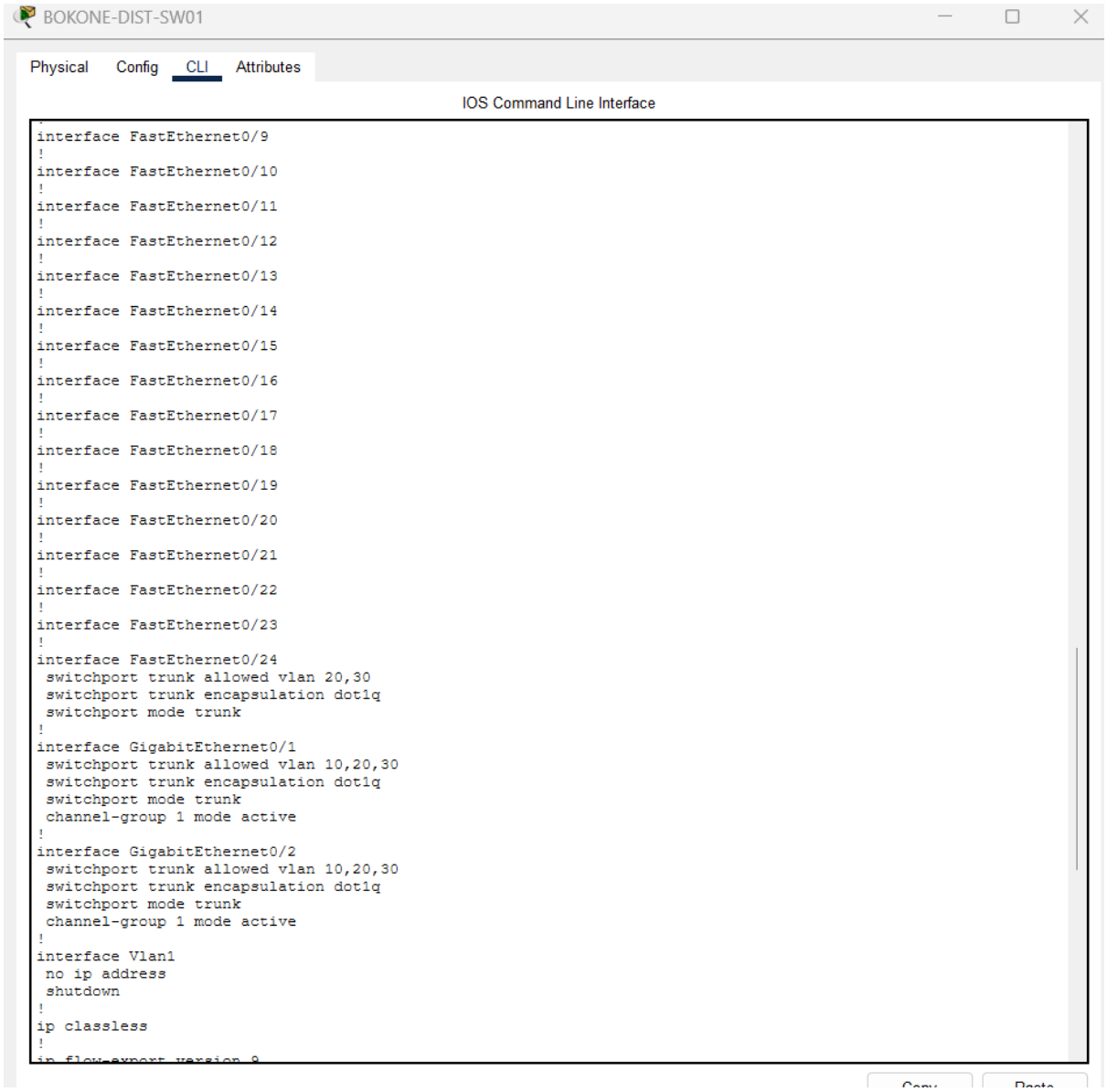
3.3 BOKONE-DIST-SW01 (Distribution Switch)

This switch forms the other end of the EtherChannel bundle and connects down to the access layer.

What was configured:

- Hostname set to BOKONE-DIST-SW01
- Same three VLANs created (10, 20, 30), matching CORE-SW01
- GigabitEthernet0/1 and GigabitEthernet0/2 configured identically to CORE-SW01's ports — trunk mode, same VLANs allowed, and bundled with channel-group 1 mode active

- FastEthernet0/24 configured as a trunk port down to the access switch, carrying VLAN 20 and VLAN 30 only (VLAN 10/MGMT is not needed at the access layer)



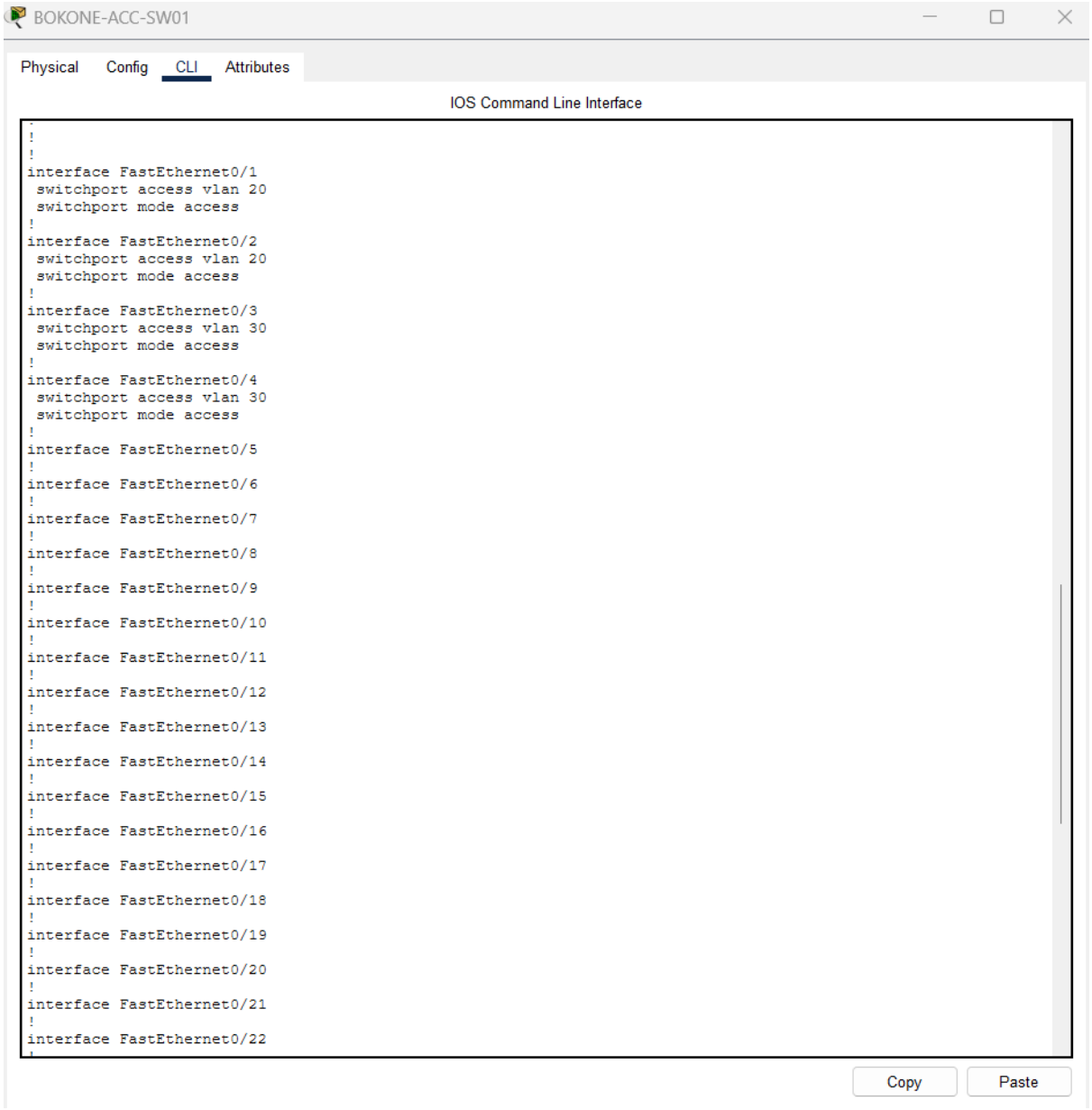
3.4 BOKONE-ACC-SW01 (Access Switch)

This is where end-user devices physically connect.

What was configured:

- Hostname set to BOKONE-ACC-SW01
- VLANs 20 and 30 created
- GigabitEthernet0/1 set as a trunk port up to DIST-SW01

- FastEthernet0/1 and FastEthernet0/2 set as access ports in VLAN 20 (for PC1 and PC2, STAFF)
- FastEthernet0/3 and FastEthernet0/4 set as access ports in VLAN 30 (for PC3 and SRV1, NOC_SRV)



4. EtherChannel Verification

This section proves the EtherChannel actually formed correctly; this is the core requirement of the assigned technical challenge.

The command show etherchannel summary was run on both CORE-SW01 and DIST-SW01. A successful result shows:

- Po1(SU) — meaning Port-channel1 exists, is a Layer 2 interface, and is in use
- Both member ports (Gig0/1 and Gig0/2) marked with (P), meaning they are properly bundled into the port-channel

```
BOKONE-CORE-SW01#
BOKONE-CORE-SW01#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
```

```
BOKONE-DIST-SW01#
BOKONE-DIST-SW01#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig0/1(P) Gig0/2(P)
BOKONE-DIST-SW01#
```

5. Testing and Verification

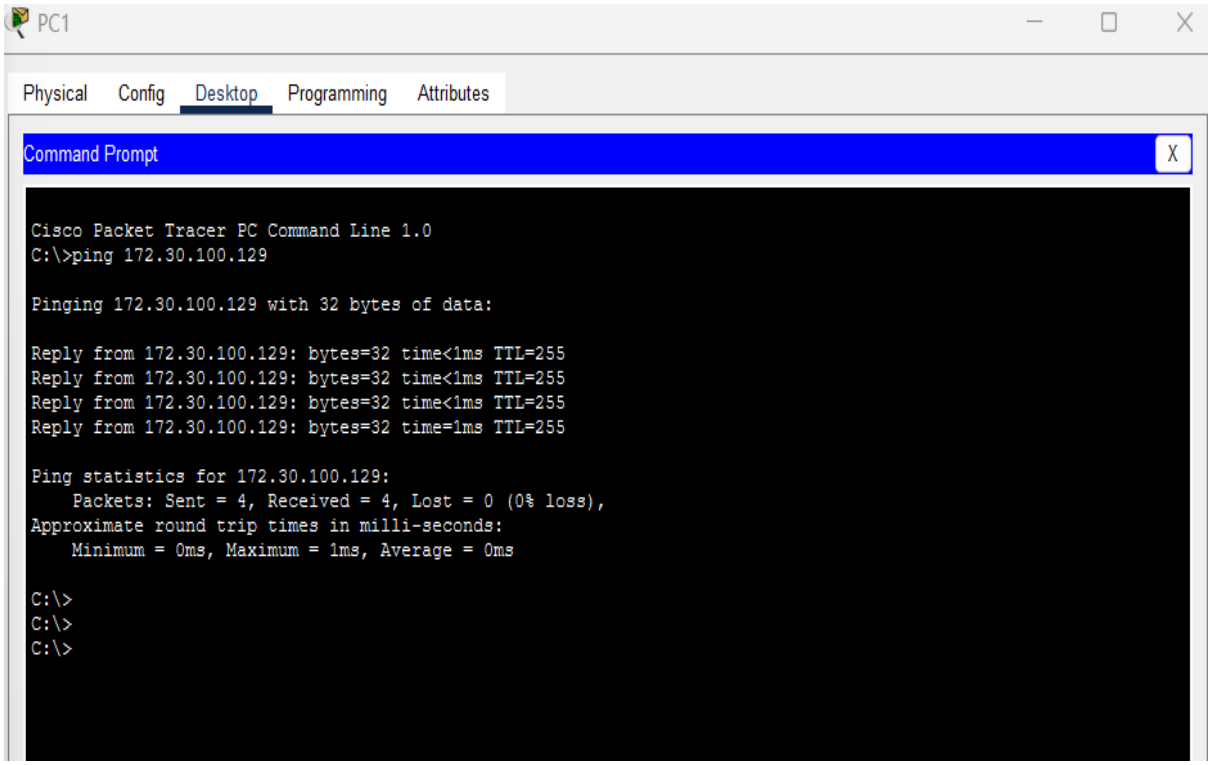
The client's requirements specifically asked for two things to be tested: normal connectivity, and what happens if one of the EtherChannel links goes down.

5.1 Test 1: Normal Connectivity

A ping was sent from PC1 (on VLAN 20, STAFF) to 172.30.100.129, the gateway address for VLAN 30 (NOC_SRV). This tests two things at once: that inter-VLAN routing works,

and that traffic is successfully crossing the EtherChannel bundle between the two switches.

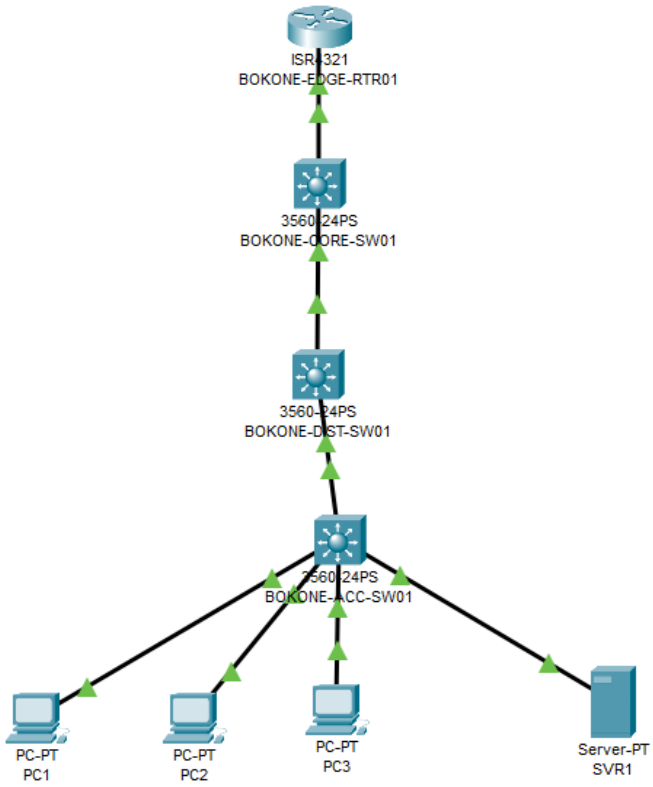
Result: 4 out of 4 packets received, 0% loss.



5.2 Test 2: Link Failure (Redundancy Test)

To prove the EtherChannel provides real redundancy, one of the two physical cables between CORE-SW01 and DIST-SW01 was deliberately removed while the network was running. The same ping was then repeated from PC1.

Result: 4 out of 4 packets still received, 0% loss, confirming that traffic automatically continued over the one remaining link, with no manual reconfiguration needed.




```
C:\>
C:\>
C:\>ping 172.30.100.129

Pinging 172.30.100.129 with 32 bytes of data:

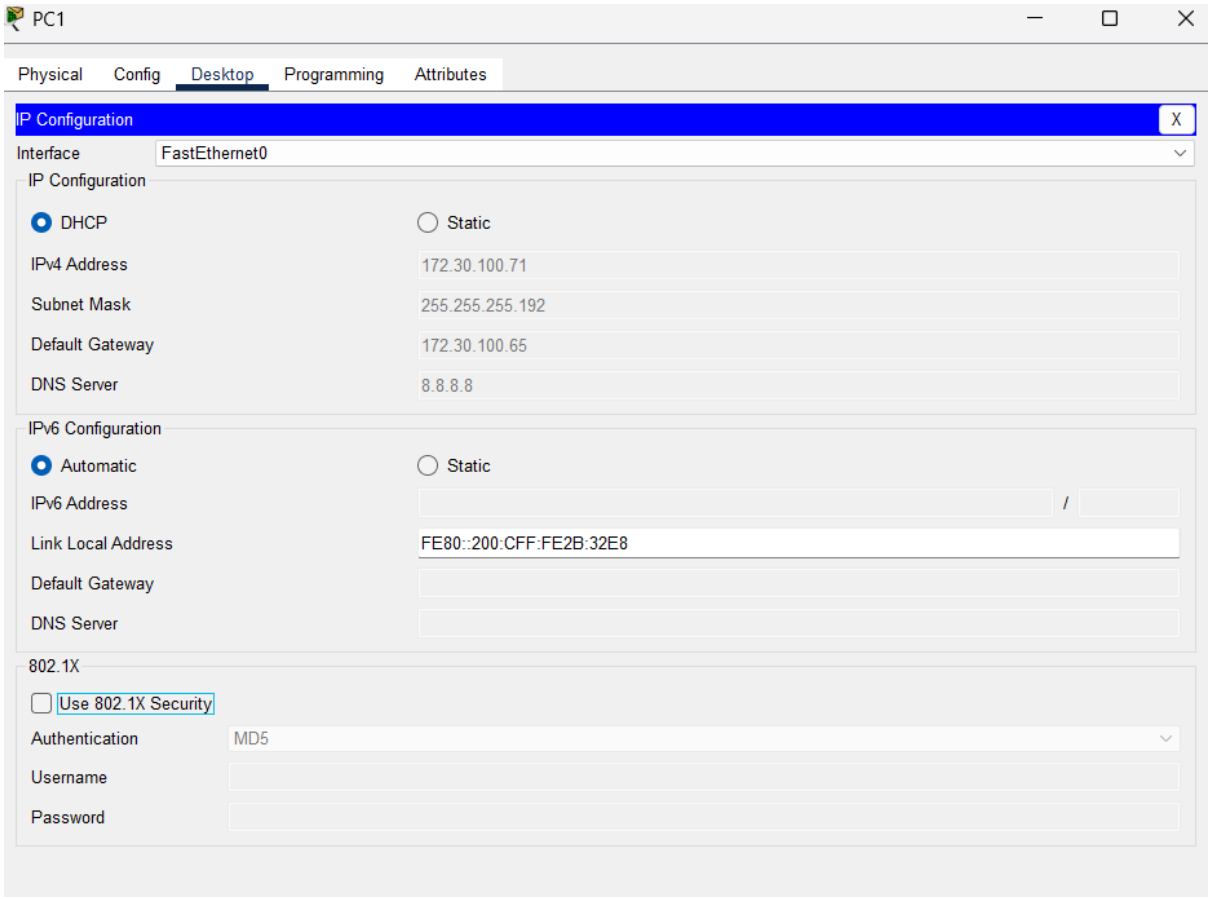
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time=10ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time=1ms TTL=255

Ping statistics for 172.30.100.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

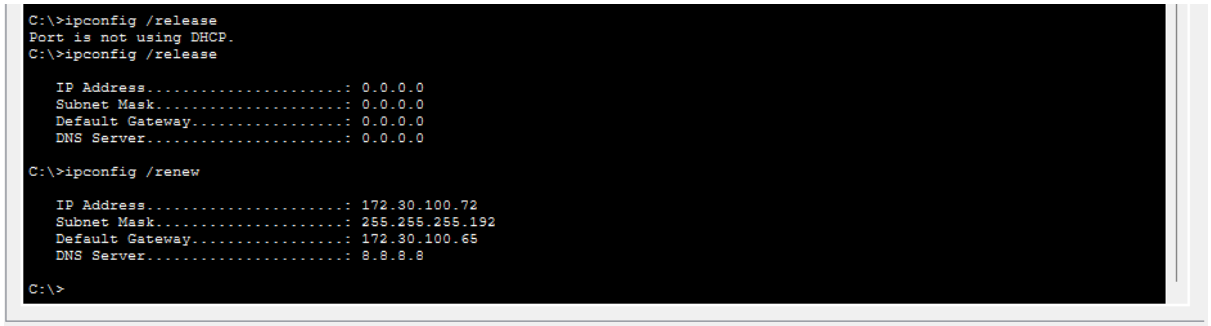
C:\>
```

5.2 Test 2: DHCP Verification

Since VLAN 20 (STAFF) is configured for DHCP rather than static IPs, this needs to be proven separately. PC1 and PC2 were checked to confirm they received an IP address automatically from CORE-SW01's DHCP pool, without manual configuration.



PC2



5.3 Test 3: DHCP Verification

VLAN 20 (STAFF) is configured to assign IP addresses automatically via DHCP, rather than manually. To confirm the DHCP service is working correctly, the command show ip dhcp binding was run on BOKONE-CORE-SW01.

Result: The DHCP server successfully issued the address 172.30.100.71 to a client device automatically, confirming the DHCP pool for VLAN 20 is active and functioning correctly.

```
BOKONE-CORE-SW01>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
172.30.100.71    0000.0C2B.32E8          --                      Automatic
BOKONE-CORE-SW01>
```

5.4 Test :Routing Path Verification

To confirm traffic between VLANs is actually being routed through BOKONE-CORE-SW01 (not just reaching the destination by coincidence), a traceroute was run from PC1 to the VLAN 30 gateway.

Command: tracert 172.30.100.129

Result: the trace completed in a single hop, confirming CORE-SW01 is directly performing the routing between VLANs as designed.

```
Pinging 172.30.100.129 with 32 bytes of data:

Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255

Ping statistics for 172.30.100.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 172.30.100.129

Tracing route to 172.30.100.129 over a maximum of 30 hops:

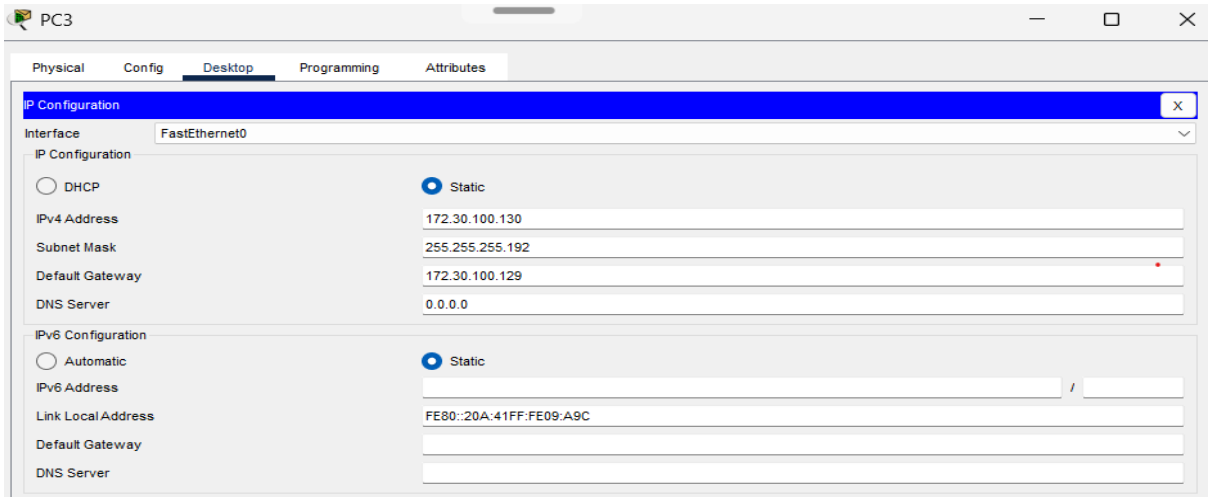
  1    0 ms      6 ms      0 ms      172.30.100.129

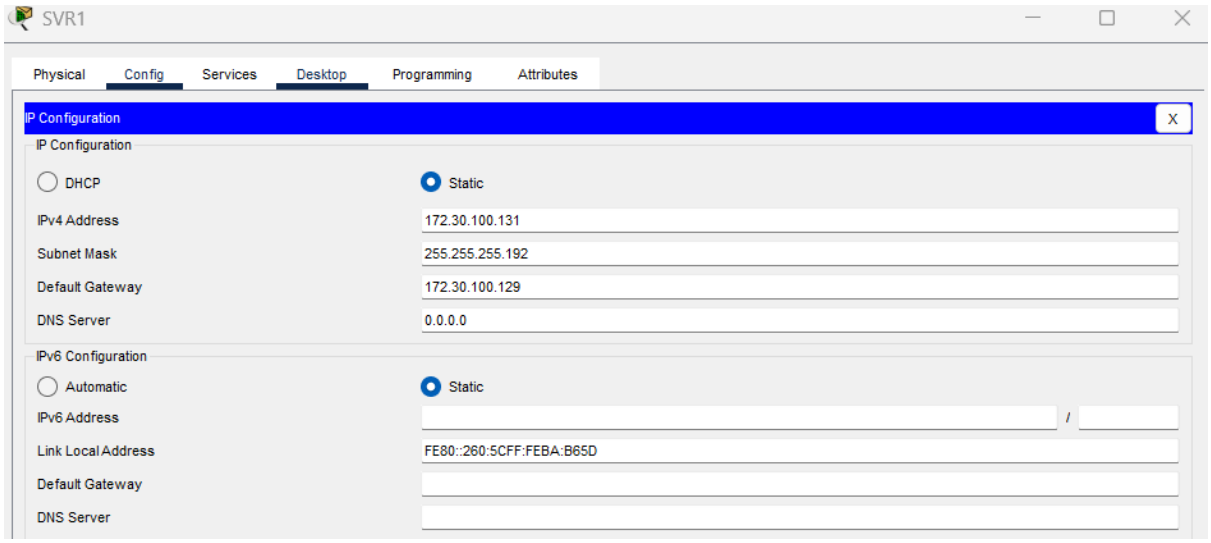
Trace complete.
```

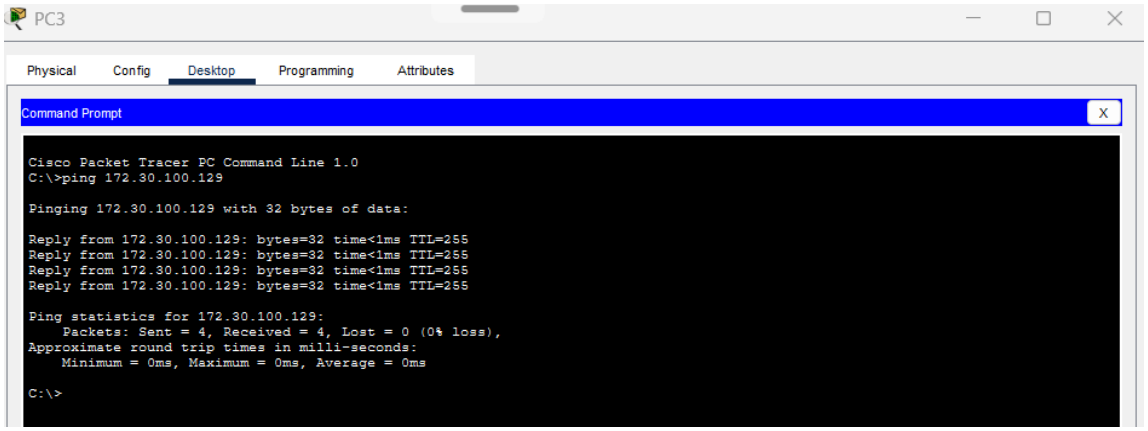
5.5 Test5: Static Addressing (VLAN 30: PC3 and SRV1)

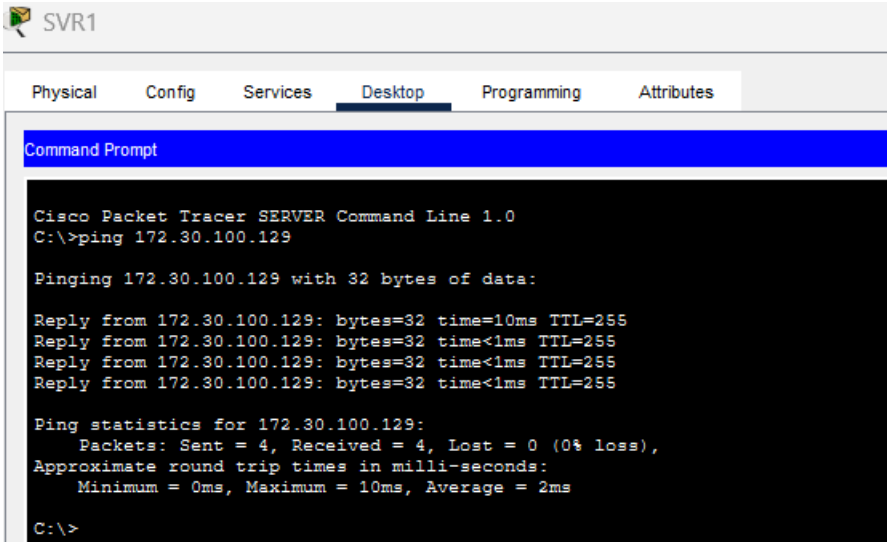
Unlike VLAN 20, which uses DHCP, VLAN 30 (NOC_SRV) uses static IP addressing. This is intentional, since it hosts the server and network operations devices that need fixed, predictable addresses rather than ones that could change.

PC3 and SRV1 were both configured with static IPs within the 172.30.100.128/26 range, with gateway 172.30.100.129. Since Test 1 (Normal Connectivity) already confirms a device successfully reaching across to that same VLAN 30 gateway and back, the same routing and connectivity logic applies here; the static devices function identically to the DHCP ones once addressed, just without the automatic assignment step.


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.30.100.130
Subnet Mask
255.255.255.192
Default Gateway
172.30.100.129
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20A:41FF:FE09:A9C
Default Gateway
DNS Server


SVR1
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
172.30.100.131
Subnet Mask
255.255.255.192
Default Gateway
172.30.100.129
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:5CFF:FEBA:B65D
Default Gateway
DNS Server


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.100.129
Pinging 172.30.100.129 with 32 bytes of data:
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Reply from 172.30.100.129: bytes=32 time<1ms TTL=255
Ping statistics for 172.30.100.129:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>



GitHub Repository

GitHub Link [CMPG325-Bokone-Community-Network]

6. Conclusion

The EtherChannel technical challenge was successfully implemented and verified. The two Gigabit links between BOKONE-CORE-SW01 and BOKONE-DIST-SW01 were bundled into a single logical Port-channel using LACP in active mode on both sides, confirmed via show etherchannel summary on both switches.

Testing confirmed the bundle performs as required:

1. **Normal connectivity**: inter-VLAN routed traffic passes correctly across the bundle
2. **Link failure:** the network continues operating with zero packet loss even when one of the two physical links is disconnected, proving genuine redundancy
3. **DHCP verification:** VLAN 20 clients correctly receive dynamic addressing from the DHCP pool, each with distinct, correctly leased IPs
4. **Routing path:** traceroute confirms traffic takes the expected single-hop path through BOKONE-CORE-SW01

VLAN 30 devices (PC3, SRV1), addressed statically per the client's requirement for fixed server/NOC addressing, were confirmed reachable using the same routed path validated in Test 1.

Together, this meets the client's requirement for a reliable, redundant switch-to-switch connection, removing the single point of failure that a single physical uplink would have created, while confirming correct VLAN segmentation, addressing, and routing across the Bokone Community Network topology.